

Candidate Name _____

Centre Number

Candidate

Number

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CAMBRIDGE INTERNATIONAL EXAMINATIONS

**General Certificate of Education Advanced Subsidiary Level
and Advanced Level**

PHYSICS

9702/2

PAPER 2 AS Core

OCTOBER/NOVEMBER SESSION 2002

1 hour

Candidates answer on the question paper.
No additional materials.

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You may lose marks if you do not show your working or if you do not use appropriate units.

FOR EXAMINER'S USE

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This question paper consists of 14 printed pages and 2 blank pages.



Data

speed of light in free space,

$$c = 3.00 \times 10^8 \text{ m s}^{-1}$$

permeability of free space,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$$

permittivity of free space,

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$$

elementary charge,

$$e = 1.60 \times 10^{-19} \text{ C}$$

the Planck constant,

$$h = 6.63 \times 10^{-34} \text{ J s}$$

unified atomic mass constant,

$$u = 1.66 \times 10^{-27} \text{ kg}$$

rest mass of electron,

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

rest mass of proton,

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

molar gas constant,

$$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

the Avogadro constant,

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

the Boltzmann constant,

$$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

gravitational constant,

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

acceleration of free fall,

$$g = 9.81 \text{ m s}^{-2}$$

Formulae

uniformly accelerated motion,

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas,

$$W = p\Delta V$$

gravitational potential,

$$\phi = -\frac{Gm}{r}$$

simple harmonic motion,

$$a = -\omega^2 x$$

velocity of particle in s.h.m.,

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

resistors in series,

$$R = R_1 + R_2 + \dots$$

resistors in parallel,

$$1/R = 1/R_1 + 1/R_2 + \dots$$

electric potential,

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series,

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel,

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor,

$$W = \frac{1}{2}QV$$

alternating current/voltage,

$$x = x_0 \sin \omega t$$

hydrostatic pressure,

$$p = \rho gh$$

pressure of an ideal gas,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

radioactive decay,

$$x = x_0 \exp(-\lambda t)$$

decay constant,

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

critical density of matter in the Universe,

$$\rho_0 = \frac{3H_0^2}{8\pi G}$$

equation of continuity,

$$Av = \text{constant}$$

Bernoulli equation (simplified),

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

Stokes' law,

$$F = Ar\eta v$$

Reynolds' number,

$$R_e = \frac{\rho v r}{\eta}$$

drag force in turbulent flow,

$$F = Br^2\rho v^2$$

Answer **all** the questions in the spaces provided.

- 1 (a) (i) Define *density*.

.....

.....

- (ii) State the base units in which density is measured.

..... [2]

- (b) The speed v of sound in a gas is given by the expression

$$v = \sqrt{\left(\frac{\gamma p}{\rho}\right)},$$

where p is the pressure of the gas of density ρ . γ is a constant.

Given that p has the base units of $\text{kg m}^{-1} \text{s}^{-2}$, show that the constant γ has no unit. [3]

- 2 A student uses a metre rule to measure the length of an elastic band before and after stretching it.

The lengths are recorded as

length of band before stretching, $L_0 = 50.0 \pm 0.1 \text{ cm}$

length of band after stretching, $L_S = 51.6 \pm 0.1 \text{ cm}$.

Determine

- (a) the change in length ($L_S - L_0$), quoting your answer with its uncertainty,

$(L_S - L_0) = \dots\dots\dots \text{ cm} \quad [1]$

(b) the fractional change in length, $\frac{(L_S - L_0)}{L_0}$,

fractional change = [1]

(c) the uncertainty in your answer in (b).

uncertainty = [3]

- 3 A ball falls from rest onto a flat horizontal surface. Fig. 3.1 shows the variation with time t of the velocity v of the ball as it approaches and rebounds from the surface.

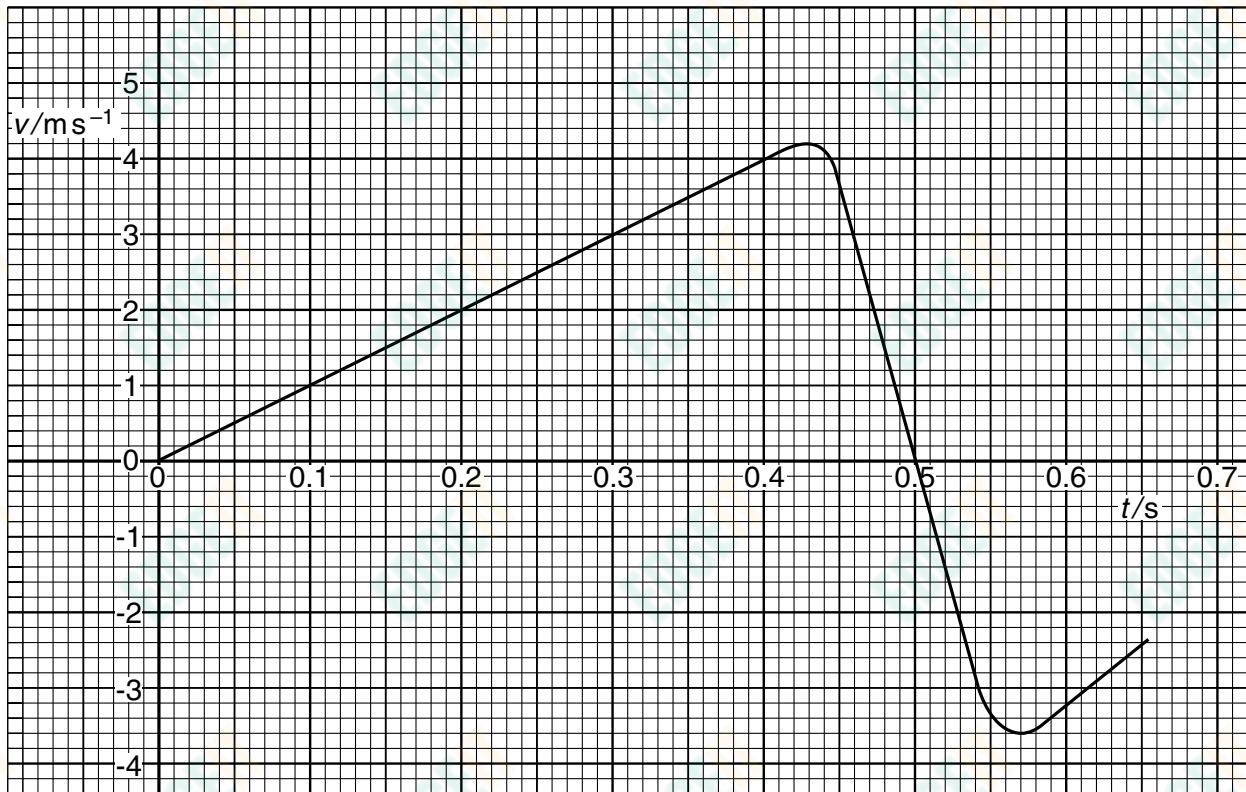


Fig. 3.1

Use data from Fig. 3.1 to determine

- (a) the distance travelled by the ball during the first 0.40 s,

distance = m [2]

- (b) the change in momentum of the ball, of mass 45 g, during contact of the ball with the surface,

change = N s [4]

- (c) the average force acting on the ball during contact with the surface.

force = N [2]

- 4 (a) Explain what is meant by the concept of *work*.

.....

 [2]

- (b) Using your answer to (a), derive an expression for the increase in gravitational potential energy ΔE_p when an object of mass m is raised vertically through a distance Δh near the Earth's surface.

The acceleration of free fall near the Earth's surface is g . [2]

- 5 The variation with time t of the displacement x of a point in a transverse wave T_1 is shown in Fig. 5.1.

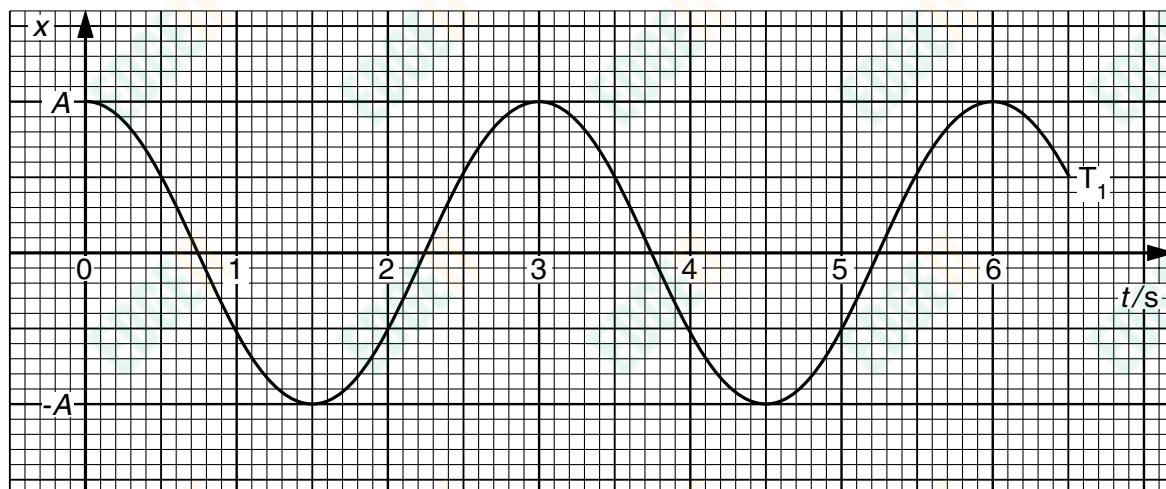


Fig. 5.1

- (a) By reference to displacement and direction of travel of wave energy, explain what is meant by a *transverse wave*.

.....
[1]

- (b) A second transverse wave T_2 , of amplitude A has the same waveform as wave T_1 but lags behind T_1 by a phase angle of 60° . The two waves T_1 and T_2 pass through the same point.

- (i) On Fig. 5.1, draw the variation with time t of the displacement x of the point in wave T_2 . [2]

- (ii) Explain what is meant by the *principle of superposition* of two waves.

.....

[2]

- (iii) For the time $t = 1.0$ s, use Fig. 5.1 to determine, in terms of A ,

1. the displacement due to wave T_1 alone,

displacement =

2. the displacement due to wave T_2 alone,

displacement =

3. the resultant displacement due to both waves.

displacement =

[3]

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Turn over for question 6

- 6 An electron travelling horizontally in a vacuum enters the region between two horizontal metal plates, as shown in Fig. 6.1.

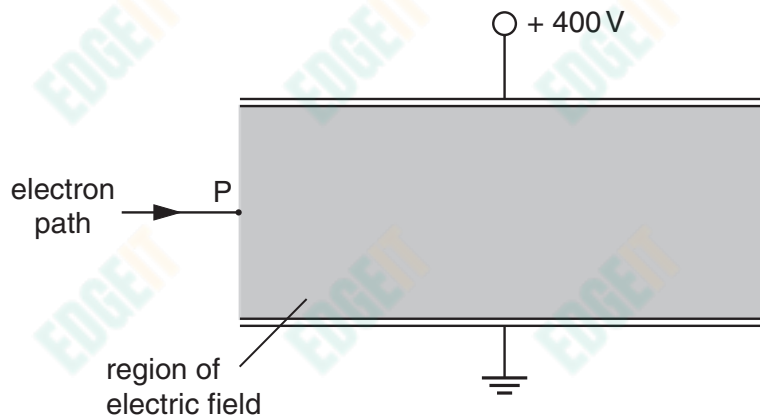


Fig. 6.1

The lower plate is earthed and the upper plate is at a potential of + 400 V. The separation of the plates is 0.80 cm.

The electric field between the plates may be assumed to be uniform and outside the plates to be zero.

- (a) On Fig. 6.1,
- draw an arrow at P to show the direction of the force on the electron due to the electric field between the plates,
 - sketch the path of the electron as it passes between the plates and beyond them.
- [3]
- (b) Determine the electric field strength E between the plates.

$$E = \dots\dots\dots \text{ V m}^{-1} \quad [2]$$

(c) Calculate, for the electron between the plates, the magnitude of

(i) the force on the electron,

force = N

(ii) its acceleration.

acceleration = m s^{-2}
[4]

(d) State and explain the effect, if any, of this electric field on the horizontal component of the motion of the electron.

.....
.....
.....[2]

- 7 A student set up the circuit shown in Fig. 7.1.

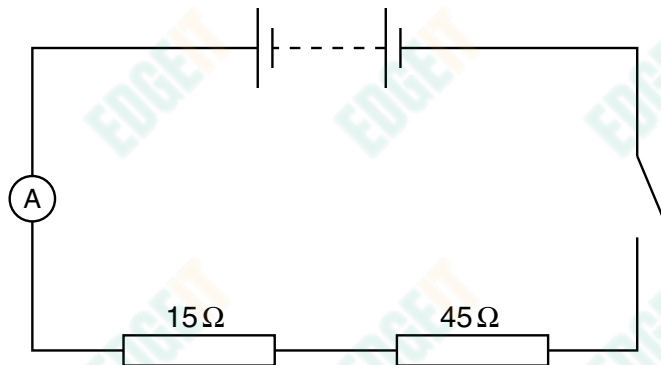


Fig. 7.1

The resistors are of resistance $15\ \Omega$ and $45\ \Omega$. The battery is found to provide $1.6 \times 10^5\text{ J}$ of electrical energy when a charge of $1.8 \times 10^4\text{ C}$ passes through the ammeter in a time of $1.3 \times 10^5\text{ s}$.

- (a) Determine

- (i) the electromotive force (e.m.f.) of the battery,

e.m.f. = V

- (ii) the average current in the circuit.

current = A
[4]

- (b) During the time for which the charge is moving, $1.1 \times 10^5 \text{ J}$ of energy is dissipated in the 45Ω resistor.

- (i) Determine the energy dissipated in the 15Ω resistor during the same time.

energy = J

- (ii) Suggest why the total energy provided is greater than that dissipated in the two resistors.

.....

.....

[4]

- 8 A nucleus of an atom of francium (Fr) contains 87 protons and 133 neutrons.

- (a) Write down the notation for this nuclide.

.....
Fr

[2]

- (b) The nucleus decays by the emission of an α -particle to become a nucleus of astatine (At).

Write down a nuclear equation to represent this decay.

[2]

- 9 An aluminium wire of length 1.8 m and area of cross-section $1.7 \times 10^{-6} \text{ m}^2$ has one end fixed to a rigid support. A small weight hangs from the free end, as illustrated in Fig. 9.1.

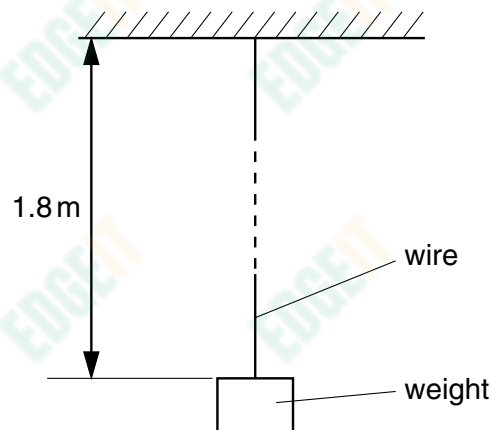


Fig. 9.1

The resistance of the wire is $0.030 \, \Omega$ and the Young modulus of aluminium is $7.1 \times 10^{10} \text{ Pa}$.

The load on the wire is increased by 25 N.

(a) Calculate

- (i) the increase in stress,

increase = Pa

- (ii) the change in length of the wire.

change = m
[4]

- (b) Assuming that the area of cross-section of the wire does not change when the load is increased, determine the change in resistance of the wire.

change = Ω [3]

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